

Calibrated microwave conversion-amplitude response to pump strength

The native HDF 5 record combines measured microwave experiments with other numerical and visualization products. This task uses the 13-mode pure frequency-conversion experiment and its 51 pump settings. Every directed off-diagonal pair is retained; complete pump matrices are the validation units. Three fixed lower settings supply the same per-pair calibration to every model.

The development-selected six-term response control reaches 3.758×10^{-5} fsu reserved MAE, compared with 4.034×10^{-5} for first-order extrapolation and 6.691×10^{-5} for holding the last calibration. This is a modest calibrated forecasting improvement. Five restricted physical explanations - shared saturation, coherent rotation, detuning-dependent saturation, incoherent power mixing and pump-induced loss - did not improve development transfer.

Let $r=(g-g_{lo})/(g_{hi}-g_{lo})$, and let L, H, N denote measured lower, higher and zero-pump response magnitudes. The reference combines L, H, $(H-L)r$, $(H-L)r^2$, N and frequency-bin separation times $(H-L)r$ with six fitted global coefficients. This transparent predictor does not establish a unique physical mechanism.

The endpoint is the magnitude of native instrument response in full-scale units. It is not an independently normalized scattering coefficient. Phase, passivity, reciprocity and quantum fidelity are outside this task, and the source paper’s existing inverse-design results are disclosed as prior art.

Experimental scope and evaluation

Predict conversion response above the two calibration pump strengths; each directed mode pair is calibrated at 0,.13,.1402040816 pump units. Whole stronger pump matrices are reserved, not selected edges.

Calibration: Native response magnitudes at three fixed low/zero-pump conditions per directed mode pair. No high-pump outcome or target standard-deviation is a predictor.

Validation unit: Complete pump-strength matrix; all 156 off-diagonal directed pairs linked. One instrument/day; strength levels are not independent devices.

Report entire pump matrix errors and extrema; no independent-device confidence interval.

Target: magnitude of native complex USB off-diagonal response (fsu; instrument full-scale units). Errors use fsu; instrument full-scale units. The selected reference is **flexible**. Selection was frozen before confirmation; diagnostic winners are not substituted afterward.

Predictor	Unit
lo	fsu; low-pump amplitude response
hi	fsu; second calibration response
noise	fsu; zero-pump response
r	dimensionless normalized pump increment
g	fsu; pump amplitude control
glo	fsu; low pump control
ghi	fsu; high calibration pump control
sep	integer frequency-bin separation; 100 kHz bins

Matched comparison

Model	Development MAE	Reserved MAE	Reserved worst-group MAE
flexible	1.53321e-05	3.75775e-05	4.82525e-05
hold_calibration	2.69472e-05	6.69148e-05	7.79037e-05
linear_increment	1.64086e-05	4.03412e-05	5.13983e-05
saturating_increment	1.65034e-05	4.10959e-05	5.22937e-05
coherent_rotation	1.64722e-05	4.1091e-05	5.22994e-05
detuning_saturation	1.6549e-05	4.11822e-05	5.19933e-05
incoherent_power_mixing	1.70114e-05	3.90719e-05	4.91777e-05
pump_induced_loss	1.64472e-05	4.10921e-05	5.23002e-05

The primary error is mean group MAE. RMSE and signed bias are complementary; rows within a group do not establish independent replication. No accuracy percentage or industrial tolerance is invented.

Selected equation and coefficients

$$\hat{A} = b_0L + b_1H + (H - L)(b_2r + b_3r^2 + b_5sr) + b_4N_0$$

L and H are the two permitted calibrated magnitudes; r is normalized pump increment, s is mode separation, and N0 is the zero-pump magnitude (the implementation calls this input noise). These are instrument full-scale units, not a quantum-noise calibration.

Coefficient	Frozen value
b0	1.19924824
b1	-0.227625824
b2	0.423759771
b3	0.109658285
b4	0.0972934482
b5	0.0294952339

All comparator expressions, numerical guards and full-precision values remain in EQUATIONS.md, rules.json and run.py.

Applicability and limitations

- Magnitude in native fsu is not a calibrated dimensionless scattering coefficient; no passivity, quantum fidelity or reciprocity claim is inferred.
- Only the pure frequency conversion 13 mode experiment is targeted.31 mode parametric gain and numerical inverse-problem products remain context, not duplicate independent observations.
- Final ten higher-strength settings test within-sweep extrapolation; no hardware transfer.

Schema audit accidentally printed source USB standard-deviation arrays; measured complex USB targets remained unprinted. Uncertainty arrays are excluded from predictors and selection.

Reproduction

Run `python run.py` in this final package to verify hashes and reproduce every saved prediction. Use `python run.py --inputs new.csv --output predictions.csv --model reference` only with the declared inputs and units. Numerical prediction evaluation does not certify novelty or mechanism. Source anchors and reserved observations are separate from predictor inputs.

Source: <https://zenodo.org/records/18938314>. See source/units audit and prior-art records in research history.